TELOMY

# Notifications & alerting, end to end.

Every notification each role — **member, doctor, diagnostic centre** — should receive, across in-app, push, email, SMS and WhatsApp; the inbound report flows (email, EML, CSV, HL7/FHIR) that trigger them; and a patient-safety critical-value path — designed India-first, DPDP-governed, and shaped to the single-worker Oracle box we actually run.

**3 roles**
user · doctor · centre

**5 channels**
in-app · push · email · SMS · WhatsApp

**Critical-value**
closed-loop safety path

**DPDP-native**
consent + quiet hours

Deep-research specification — audit, benchmarks, full matrix, inbound flows, architecture & roadmap. Prepared 28 Oct 2026. India regulatory and competitor claims are from secondary sources — verify against primary before build or public use.

# Part A — What exists today

Telomy has **one** notification surface: a pull-model, in-app-only list. No stored record, no delivery, no channel beyond the app screen.

**The list that exists**
`features.py:159 notifications()` recomputes on every request — 6 patient kinds: **shared_doc, retest, review, insight, appointment, medication**. Served at `/notifications`; read-state is a dedup set only (`notif_read`, key `kind:title`) — no row, timestamp, channel or audit.

**Channels: in-app only**
Exhaustive grep for any email/SMS/WhatsApp/push sender returns **zero**. No `expo-notifications`, no push tokens. OTP is **stubbed** (`000000`). Appointment booking says "confirms by SMS" but **sends nothing**.

**Infra exists but inert**
`interop.py` has outbound webhooks (HMAC-signed `fire()`) + FHIR R4 export — but `fire()` has **no callers**. The "notify an external EHR on a result" plumbing is built and never triggered.

**Consent yes, prefs no**
8 DPDP purposes modeled (incl. a separate **marketing** opt-in). But **no per-category / per-channel / quiet-hours** preference model. Staff side has coarse booleans with no delivery behind them.

**Inbound = PDF only**
Lab reports parse from PDF (`labparse.py`). **No EML / email / CSV / inbound-HL7** path. Device inbound (WHOOP, CGM, patch) is telemetry, not reports.

**No safety tier**
Marker status is whatever the client sends or "unknown". **No critical/panic tier, no auto-flag on ingest, no safety alert.**

## BUILT VS MISSING

| CAPABILITY | STATUS | CAPABILITY | STATUS |
|---|---|---|---|
| In-app list (6 kinds) | **Built** | Push (APNs/FCM/Expo) | **Missing** |
| Read-state | **Partial** (dedup set) | Email transactional | **Missing** |
| Stored timestamped records | **Missing** | SMS (DLT) | **Missing** (copy only) |
| DPDP consent model | **Built** | WhatsApp (BSP) | **Missing** |
| Per-channel prefs + quiet hours | **Missing** | Critical-value tier | **Missing** |
| Outbound webhooks / FHIR export | **Built, inert** | EML/CSV/HL7 inbound | **Missing** |
| Doctor/centre push alerts | **Missing** (polled) | Client offline outbox | **Built** (uploads) |

**Infra constraint shaping everything:** single `uvicorn --workers 1` on an Oracle micro (≤1GB), no queue/worker, WAL SQLite per-user. Generation & digests must be **cron-driven batch**; sends must be cheap, rate-limited, provider-outsourced — never a resident worker pool.

# Part B — Benchmarks & India rules

Secondary sources (vendor docs, law-firm/BSP summaries). Verify DLT/WhatsApp specifics on the live Meta & operator portals, and DPDP against the notified Rules, before build.

### Consumer labs (Function / Superpower)

Every value flagged in/out of range; **clinician review framed as "talk to a clinician," never a diagnosis**. Results come with context + a next step, never a bare "you're abnormal" push — the pattern our `clinician_notes` + review queue already mirrors.

### Wearables (Oura / Whoop)

One **morning** Readiness/Recovery push; anomaly nudges (elevated resting HR, temp); missing-night & sync reminders; streaks. Batched to a daily summary, not per-metric spam. We already compute the equivalents — just don't deliver them.

### India channel strategy

WhatsApp is a **primary** report/appointment channel (Apollo 24|7, Tata 1mg, Practo, Eka Care) alongside SMS + push, because open rates dwarf email. Stack: **SMS (DLT) for OTP/must-deliver, WhatsApp utility for reports/appointments, push for engagement, email for records/PDFs.**

### Critical-value alerting (safety)

Accreditor norm (CAP / Joint Commission, secondary): **≥90% of critical values reported within ~30 min**; closed-loop **read-back** documented; recipient a **licensed caregiver**; an **escalation policy** for non-ack.

## INDIA REGULATORY — MUST HONOR

| REGIME | WHAT IT REQUIRES |
|---|---|
| **DPDP Act 2023 + Rules** (notified Nov 2025) | Consent free/specific/informed/unambiguous, **notice precedes consent**, withdrawal as easy as giving. Health data likely **Significant Data Fiduciary** (DPO, DPIA, audits). **Medical-emergency / threat-to-life needs no consent** — the basis to let a true critical alert override an opt-out. Rule-4 phases in toward **13 Nov 2026**. |
| **TRAI TCCCPR / DLT** (SMS) | Register **entity + header + content template**; carriers block anything unmatched. **Transactional & Service-Implicit reach DND 24×7; Promotional needs explicit opt-in, 09:00–21:00 IST only.** |
| **WhatsApp Business Platform** | Templates categorized **Utility / Authentication / Marketing**; promo content in a utility template gets reclassified (higher billing). **Opt-in required**; a user-initiated **24h service window** allows free-form replies. Auth templates = OTP only, no links. |
| **Email deliverability** | Publish **SPF, DKIM, DMARC**; separate transactional vs marketing streams/subdomains. |

# Part C — The notification matrix

**Priority tiers & defaults:** CRITICAL push+SMS+WhatsApp, bypass quiet hours, closed-loop ack, vital-interest basis — never the clinical value on a lock screen. HIGH push primary, WhatsApp fallback if unread, quiet hours except time-sensitive. NORMAL in-app+push, batched, quiet hours honored. LOW opt-in engagement, digest, quiet 21:00–09:00 IST, unsubscribe. Every category has a per-channel toggle except safety and OTP/security (always on).

## C.1 — MEMBER / PATIENT

| # | NOTIFICATION | TRIGGER | CHANNELS | PRIORITY |
|---|---|---|---|---|
| U1 | Lab result ready | Report ingested, markers persisted | Push + in-app; WA fallback; email+PDF | NORMAL |
| U2 | New non-lab document (imaging, discharge, Rx) | `/documents/parse` or share lands | In-app + push | NORMAL |
| U3 | Abnormal value flagged | Ingest marks marker out-of-range | Push + in-app; WA — "review with Sinc/clinician" | HIGH |
| U4 | **Critical / panic value** | Ingest hits critical threshold ($K^+$>6, glucose>400, Hb<7…) | **Push+SMS+WA, closed-loop ack; alerts panel doctor** | CRITICAL |
| U5 | Telomy Score / bio-age changed | Recompute crosses materiality delta | In-app + push (into morning digest) | NORMAL |
| U6 | Retest due | `retest.plan` crosses due date | In-app + push; WA if long-overdue | NORMAL → HIGH |
| U7 | "What changed since last time" | Retest ingested, marker improved/worsened | In-app + push | NORMAL / HIGH |
| U8 | Doctor shared a doc — confirm to add | `sharedocs.add()` → pending | In-app + push; WA | HIGH |
| U9 | Doctor message / consult recap ready | Scribe recap or secure message | In-app + push; WA (link, not detail) | HIGH |
| U10 | Finding signed / care-plan updated | Doctor signs off, plan approved | In-app + push | NORMAL |
| U11 | Sinc finding awaiting attention | Insight ≥0.75 or sent for review | In-app + push | NORMAL |
| U12 | Appointment reminders | Upcoming appointment | In-app+push; **SMS+WA at 24h & 2h** | HIGH |
| U13 | Appointment booked/cancelled/rescheduled | Status change | In-app + push; WA | HIGH |
| U14 | Medication / supplement dose reminder | Dose due today & untaken | In-app + push (local scheduled) | NORMAL |
| U15 | Adherence slipping | 30-day adherence below threshold | In-app + push (weekly digest) | LOW |
| U16 | Therapy session / protocol step due | Therapy/protocol schedule | In-app + push | NORMAL |
| U17 | Drug-interaction / caution flag | Interaction match on active meds+events | In-app + push — "check with clinician" | HIGH |
| U18–21 | Device not syncing / low battery / night missing / capture incomplete | Signal gaps, BLE battery | In-app (+push opt-in), debounced, morning | LOW |
| U22 | Daily morning summary | Cron each morning (absorbs U5/U6/U20/U23) | Push + in-app | LOW |
| U23–25 | Streak milestone / weekly-monthly report / education | Stats, report gen, content | In-app + push + email | LOW |
| U26 | OTP / login code | `/auth/otp/send` | **SMS (DLT) + WA auth template** | HIGH |
| U27–28 | New-device login / password change | Auth events | Push + email; SMS if high-risk | HIGH |
| U29 | Consent request / change confirmation | New purpose or consent toggled | In-app + email (notice-before-consent) | HIGH |
| U30–31 | Data export ready / deletion done / vault access | Export, erasure, shares | In-app + email (secure link) | NORMAL |
| U32 | Billing (ok, failed, renewal, expiry) | Billing events | In-app + email; SMS/WA on failure | HIGH / NORMAL |

## C.2 — DOCTOR

| # | NOTIFICATION | TRIGGER | CHANNELS | PRIORITY |
|---|---|---|---|---|
| D1 | **Critical value on a panel patient** | Ingest flags critical for a linked patient | **Push+SMS+WA, closed-loop ack, escalate to centre if unacked** | CRITICAL |
| D2 | New patient assigned / linked | Link or centre assignment | In-app + push | NORMAL |

| | | | | |
|---|---|---|---|---|
| D3 | Patient confirmed / declined a shared doc | `sharedocs.set_status` | In-app + push | NORMAL |
| D4 | Finding awaiting sign-off | Insight → `awaiting` | In-app + push (gated by notify_findings) | HIGH |
| D5 | Lab result back for an ordered test | Result ingested for a panel patient | In-app + push; email digest | HIGH |
| D6 | New consult / message from patient | Message or consult request | In-app + push; WA opt-in | HIGH |
| D7 | Appointment booked / cancelled / rescheduled | Status for this doctor | In-app + push; SMS same-day | HIGH |
| D8 | Care-queue backlog threshold | Awaiting reviews exceed threshold/age | In-app + push; email daily | NORMAL |
| D9 | Co-sign / approval request | Treatment/therapy/Rx needs sign-off | In-app + push | HIGH |
| D10–12 | Patient acknowledgement / schedule reminder / monthly report awaiting | Patient acts, upcoming consult, report state | In-app + push; SMS 30-min pre-consult | NORMAL / LOW |
| D13 | Account / security | Staff auth events | Push + email (always on) | HIGH |

## C.3 — CENTRE / CLINIC ADMIN

| # | NOTIFICATION | TRIGGER | CHANNELS | PRIORITY |
|---|---|---|---|---|
| C1 | **Critical-result escalation** | Patient's critical value unacked by doctor within SLA | **Push+SMS+WA to on-call admin** | CRITICAL |
| C2–3 | New doctor onboarded / new patient | Staff or member added | In-app + email | NORMAL / LOW |
| C4 | Ops: test backlog / sample pending | Tests not collected/resulted past threshold | In-app + push; email digest | HIGH |
| C5 | **TAT breach** (turnaround-time) | Result not delivered within target TAT | In-app + push; email | HIGH |
| C6 | Booking activity & no-shows | Bookings / cancellations | In-app + push | NORMAL |
| C7 | Billing / revenue / credits low | Billing + credits near total | In-app + email; SMS for large outstanding | HIGH / NORMAL |
| C8–9 | Outcomes report / staff productivity | Period report, sign-off volume | In-app + email (weekly) | NORMAL / LOW |
| C10 | Integration / device health | Source silent, webhook failures, ingest errors | In-app + push; email | HIGH |
| C11 | **Compliance / consent / audit** | Consent revoked, DPDP erasure, unusual access_log | In-app + email | HIGH |
| C12 | Account / security (admin login, role, API key) | Auth + api_keys events | Push + email (always on) | HIGH |

## Part D — Inbound report flows

Principle: **ingest** → **normalize** → **persist** → **auto-flag** → **notify**, with the notify tier decided by the auto-flag. Today only the PDF path exists and it stops at persist.

**1 · PDF upload (built)**
`/labs/parse_b64` → confirm markers → `/labs/save`. **Add** the auto-flag step on save, then emit U1/U3/U4 (+ D5/D1 for panel patients).

**2 · Email ingestion (build)**
Per-user address `reports+<token>@in.telomy.health` + shared clinic inbox. Provider inbound route → `POST /ingest/email`. Verify SPF/DKIM + allow-list; else a review queue. MIME walk → PDF/CSV/EML/FHIR dispatch → auto-flag → U1.

**3 · EML / forwarded email (build)**
Same MIME walker, uploaded as a file via an extended `/documents/parse`. For "patient forwards the lab's email" — parse the nested `message/rfc822` and recurse.

**4 · CSV (build)**
Columns → markers map (test, value, unit, ref range, date). Bulk/historical import & lab portals. Same auto-flag + U1.

**5 · HL7 v2 / FHIR R4 inbound (build)**
We already **export** FHIR; add inbound `POST /interop/fhir` + an HL7 ORU^R01 adapter, authed by existing scoped **API keys**. The clean B2B lab-integration path.

**6 · Device APIs (built, keep distinct)**
WHOOP, cal, Nightscout CGM, vitals, Apple Health are **telemetry, not reports** — they feed U18–21 & the score, **not** U1/U4. Keep the critical tier scoped to lab markers.

### D.2 · CRITICAL-VALUE AUTO-FLAGGING (NEW, PATIENT-SAFETY)

A deterministic step on **every** ingest path before notifying: compare each marker to its reference range **and** a new **critical/panic threshold table** ($K^+$, $Na^+$, glucose, Hb, platelets, WBC, creatinine, troponin, calcium, INR). Add a `critical` status. Tier → `critical`: U4+D1(+C1); `out_of_range`: U3; else U1. Record provenance for audit + the ack log. Reuse the inert `interop.fire('lab.result')` to notify subscribed EHRs.

**Safety copy rule.** The patient notification **never** states the dangerous number on push/SMS — "an important result needs attention, open Telomy." The doctor alert carries the clinical value, because a licensed caregiver is in the loop.

# Part E — Architecture & roadmap

No queue/worker, fragile box → **do not add Celery/Redis.** Store, generate by cron, deliver by thin provider calls, keep the critical path synchronous.

**① Notification store**
Real table `notifications(…, created_at, read_at, dedup_key, channels_json, delivered_json, ack_at)`. Builders **write rows** (idempotent on dedup_key) instead of recomputing; `notif_read` → `read_at`.

**② Generation = cron**
A lightweight cron (how backups already run) scans triggers every few minutes, inserts rows + enqueues sends into a SQLite `notif_outbox`. Mirrors the mobile outbox; survives restarts.

**③ Delivery = thin calls**
A second cron drains the outbox, calling provider HTTP APIs with short timeouts (like `interop.fire`'s 5s urllib), marking delivery per channel, retrying with backoff. No persistent connections.

**④ Critical path synchronous**
U4/D1/C1 fire **inline** on ingest, open a closed-loop ack record; a cron sweep escalates unacked critical alerts (doctor → centre) per SLA.

## CHANNEL ADAPTERS

**Push** — `expo-notifications` + token registration, send via the **Expo Push API** (single HTTPS endpoint, no SDK on the box — cheapest for the micro); local scheduled for dose reminders. **Email** — SES/Postmark + SPF/DKIM/DMARC on `telomy.health`; inbound route for `/ingest/email`. **SMS** — DLT-registered Indian BSP (MSG91/Gupshup/Kaleyra), templates for OTP, reminders, critical "open app," billing failure. **WhatsApp** — Business via BSP with approved utility/auth templates + opt-in capture; 24h window for free-form consult replies.

## PRIORITIZED BUILD ROADMAP

| PHASE | SHIPS |
|---|---|
| **0 · Storage & safety** | Notification store + outbox; migrate builders to write/read. **Critical-value threshold table + auto-flag on ingest.** Device push (Expo). *Patient-safety must precede delivery so U4/D1 are correct.* |
| **1 · Must-deliver TXN** | SMS via DLT (replace the `000000` OTP stub, reminders, critical "open app," billing failure). Real appointment confirm/reminder (closes the fake "by SMS"). Email transactional. **Critical → doctor → centre escalation** with ack log. |
| **2 · India + inbound** | WhatsApp Business (utility/auth templates + opt-in). Inbound email/EML/CSV ingest → auto-flag. Preferences + quiet-hours; morning-digest batching (U22). |
| **3 · Integrations & ops** | Inbound FHIR/HL7 + activate `interop.fire()`. Centre ops: TAT field + breach, backlog, compliance/audit. Engagement/marketing behind consent. |

## OPEN RISKS TO SETTLE BEFORE BUILD

- **Patient-safety posture.** Telomy is positioned as wellness ("not a diagnosis"). Decide whether we **assert** criticality (clinical liability, needs licensed-clinician loop) or **route to a clinician** without asserting. **Recommendation:** route for patient-facing copy; assert only on the doctor/centre safety path.
- **Threshold ownership.** The critical/panic table needs clinical sign-off (age/sex-specific, India reference ranges) — not an engineering decision.
- **Box capacity.** If synchronous critical sends + digests strain the 512MB box (the known wedge), move the delivery cron to a tiny separate process / managed function; keep FastAPI doing only inline critical sends.
- **Regulatory, verify primary:** CAP/Joint-Commission timeframe & read-back wording; DPDP Rule-4 date (~13 Nov 2026) & SDF status at our scale; TRAI header-suffix scheme; WhatsApp 24h-window mechanics & template SLAs.

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